Averaging versus Aggregating Inflation Expectations

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Abstract

A large literature documents that household inflation expectations are heterogeneous and biased, and summarizes them by the equal-weighted average across survey respondents. I argue that the policy-relevant object is instead an aggregate that weights each household by its stake in the macroeconomic variable being forecast. My central result is that this stake is pinned down twice over: the consumption share is both the weight that reconciles household forecasts with the published price index and, in an otherwise standard heterogeneous-agent New Keynesian model perturbed only by heterogeneous beliefs about next-period inflation, the weight on household beliefs in the aggregate Euler equation. The weight that reconciles the survey with measured inflation is the weight that drives aggregate demand. Because lower-spending households expect higher inflation, the correctly aggregated belief lies well below the average: households look far less irrational, aggregating closes about a third of the gap to professional forecasters and roughly halves the forecast bias, and what remains is a modest level bias rather than the information rigidity that afflicts the professionals. Because the average sits above the belief that governs demand, a central bank that reads it overstates how pessimistic households are and risks tightening more than the demand-relevant expectation warrants. Other macroeconomic objects call for other stakes, labor income for the wage Phillips curve and wealth for bond pricing, but the equal-weighted average matches none of them.

Keywords: inflation expectations; household surveys; aggregation; expenditure shares; heterogeneous agents; New Keynesian model.

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1. INTRODUCTION

Macroeconomic analysis routinely summarizes the beliefs of a heterogeneous population by a single number. Survey measures of expected inflation, output growth, unemployment, and asset returns are aggregated across respondents and then compared to a realized aggregate, used as a regressor in a Phillips curve, or fed into a representative-agent model. This aggregation step is almost always performed by equal weighting across respondents, or by population post-stratification, so that each household counts in proportion to its share of the population. Yet the macroeconomic objects to which these aggregates are compared are not population averages of household-level quantities. They are structurally weighted aggregates, in which households enter in proportion to their economic size. The mismatch between how beliefs are aggregated and how the corresponding macroeconomic aggregate is constructed is, I argue, a first-order and largely overlooked source of measurement error in the empirical expectations literature.

This paper studies the mismatch for inflation, the most consequential expectation to get right. Inflation expectations are the ones central banks track most closely, the ones that enter the Phillips curve and the Euler equation that govern aggregate supply and demand, and the ones whose "anchoring" is itself an objective of monetary policy. Inflation is also the setting in which the correct aggregator can be characterized exactly: the Consumer Price Index is by definition the expenditure-share-weighted average of the price changes individual households face, and households are known to form their reported expectations through the prices they personally encounter, so the aggregation question has a precise answer rather than an approximate one. A large literature reads the household inflation survey as evidence of irrationality. Carroll [2003] shows that Michigan responses are biased upward and lag professional forecasts; Mankiw, Reis & Wolfers [2003] document disagreement absent among professionals; Coibion & Gorodnichenko [2015] find household forecast errors predictable from forecast revisions; Bhandari, Borovička & Ho [2024] rationalize the gap through an estimated belief-distortion parameter. The common reading is that household inflation expectations are too high, too dispersed, and in need of correction before they can be used in policy analysis or in structural models. I take this heterogeneity as given and ask not why it arises but how it should be aggregated.

I make three contributions. *First*, and most importantly, I construct the correct aggregator for household inflation beliefs and show that it is the object that governs aggregate demand. The accounting half is immediate: because the Bureau of Labor Statistics builds the Consumer Price Index as $\pi_t^{\mathrm{CPI}} = \sum_i \left(E_i / \sum_j E_j\right) \pi_{i,t}$, aggregate inflation is identically the expenditure-share-weighted average of household-level inflation, so the consumption-share-weighted *aggregate* of household forecasts is the combination consistent with the index, while the equal-weighted *average* the literature uses corresponds to no object the index measures; the two differ by exactly the cross-sectional covariance between expenditure and expected inflation. The structural half is the main theoretical result: in an otherwise standard heterogeneous-agent New Keynesian model into which I inject only heterogeneous beliefs about next-period inflation, the household Euler equation that determines aggregate demand loads on *precisely* this consumption-share aggregator. The weight that reconciles the survey with the published index is the very weight that drives household spending. Since the dominant reason to track household inflation expectations is their effect on demand, this coincidence is the result that matters most. The comparison is silent on whether households are right (it is a statement about weights, not accuracy), and the cross-sectional input it needs holds sharply in the data: in the Survey of Consumer Expectations mean expectations fall from 4.72 percent in the second-lowest of eleven income brackets to 2.80 percent in the top, and households earning more than $100,000 are about a quarter of the population but account for 46 percent of aggregate expenditure.

Second, once households are aggregated correctly they are considerably less pessimistic, and so look considerably less irrational, than the conventional average implies. Across surveys, consumption-side weighting lowers the household aggregate by 0.2 to 0.8 percentage points and closes between a quarter and a half of the measured gap to professional forecasters. Treating rationality as a hypothesis rather than an assumption, I subject the constructed aggregate to standard bias and efficiency tests: correct aggregation roughly halves the household forecast bias relative to the equal-weighted average, and households pass the forecast-efficiency test that the professionals fail. Read through the decomposition of full-information rational expectations in Coibion & Gorodnichenko [2015], the aggregate's residual departure is a modest level bias rather than the information rigidity that afflicts the professionals, and it is far smaller than the equal-weighted average implies, so aggregation does not by itself render household expectations fully rational but it removes the part of the apparent irrationality that was an artifact of weighting. The post-2020 surge makes the stakes concrete: the consumption-weighted household aggregate tracked realized inflation to within half a percentage point while professional forecasters under-predicted it by more than two, so in the most consequential inflation episode in four decades correct aggregation moved households toward, not away from, the realized outcome.

Third, the same model shows what the consumption aggregator is *not*, and so maps out which weight a researcher should use for other questions. The wage Phillips curve loads on the *labor-income-share*-weighted aggregate, and bond-market clearing, the equilibrium Fisher relation for a bond whose payoff is eroded by inflation, loads on the *wealth-share*-weighted aggregate, the belief of the marginal bondholder. Each is meaningfully below the equal-weighted average, and they differ from one another: the wealth aggregator, which downweights the asset-poor high-expectation households most aggressively, implies the largest correction. These are the right objects for studying wage dynamics or bond pricing, but they are secondary to the demand channel of the first contribution; what unifies them is that the equal-weighted average the literature feeds into Phillips curves and policy deliberations corresponds to none of these structural objects.

With the three aggregators in view, the paper's place in the literature is clear. The cross-sectional heterogeneity that drives the wedge is the subject of a large recent body of work: lower-income and lower-education households report higher and more dispersed inflation expectations [D'Acunto, Malmendier & Weber, 2023, Weber et al., 2022], shaped by grocery-price exposure [D'Acunto et al., 2021], lifetime experience [Pedemonte, Toma & Verdugo, 2023], and personal price experiences [Kaplan &

Schulhofer-Wohl, 2017]. I take that heterogeneity as given and ask how to weight it; the same work supports the consumption-share weight on the demand side, because the spending of higher-income, financially unconstrained households responds more strongly to inflation expectations [Weber et al., 2022, Coibion et al., 2023, Bachmann, Berg & Sims, 2015], exactly the households the aggregate Euler equation up-weights. A second strand studies how expectations form and depart from full-information rational expectations, through individual over-reaction with consensus under-reaction [Bordalo et al., 2020], information rigidity [Coibion & Gorodnichenko, 2015], the under- then over-shooting of Angeletos, Huo & Sastry [2021], and the difficulty of sustaining rational expectations about equilibrium prices in heterogeneous-agent models [Moll, 2025]. My exercise is orthogonal to this question: I take the cross-section of beliefs as given, whatever its origin, and the rationality tests of Section 6.4 connect the resulting aggregate to that literature.

A distinct, methodological strand asks how much survey expectations can teach us at all. After the rational-expectations revolution dismissed elicited beliefs as noise, recent work has rehabilitated them, showing that survey expectations are meaningful and that they predict households' consumption, saving, and portfolio decisions [D'Acunto & Weber, 2024]; the evidence is collected in the Bachmann, Topa & van der Klaauw [2023]. Adam, Matveev & Nagel [2021] establish that survey expectations are subjective physical beliefs rather than risk-neutral or risk-adjusted forecasts, which is precisely what licenses reading a survey response as the belief $E_i^t[\pi_{t+1}]$ that enters each household's Euler equation and the bond-pricing condition of my model. I contribute to this agenda from the aggregation side: I expand what survey expectations can do by aggregating them correctly. The same responses become more useful, and less irrational, once they are combined with the structurally appropriate weights, because much of the pessimism that fueled their dismissal is an artifact of equal-weighting rather than a property of the beliefs themselves.

Two recent papers are especially close. Reis [2021] studies the discrepancy between survey ("people") and market measures of inflation expectations, with the market measure set by the belief of the marginal trader rather than any average; the consumption- and labor-income-share aggregators here refine the "people" side, while the wealth-share aggregator is the frictionless benchmark for the "market" side, to which the marginal investor's belief collapses absent portfolio constraints. Bhandari, Borovička & Ho [2024] feed survey pessimism into a business-cycle model as a structural belief distortion that depresses aggregate demand; I show that a meaningful part of the inflation pessimism relevant for demand is an artifact of equal-weighting rather than a distortion of the demand-relevant belief. The across-respondent aggregation studied here is in turn distinct from, and complementary to, the within-respondent aggregation of Dietrich et al. [2023].

The remainder of the paper develops the accounting framework (Section 2), generalizes a standard heterogeneous-agent New Keynesian model to heterogeneous inflation beliefs and derives the object-dependent aggregators, with the consumption-share demand aggregator as the central result (Section 3), and describes the data (Section 4). The empirical analysis then documents that inflation expectations decline with every economic stake (Section 5), and compares averaging with aggregating: the structural aggregates are significantly less pessimistic than the equal-weighted average, they close much of the gap to professional forecasters, before subjecting the aggregate to rationality tests (Section 6). Section 7 concludes.

2. ACCOUNTING FRAMEWORK

This section derives the aggregator of household inflation expectations that is consistent with the structure of the Consumer Price Index. I work with a continuum of households indexed by $i \in [0,1]$ and a finite collection of commodity categories indexed by $k \in \{1,\ldots,K\}$.

2.1. Aggregator for inflation

The Consumer Price Index is a Laspeyres fixed-weight price index,

$$\mathrm{CPI}_t = \sum_{k=1}^{K} \omega_k \, p_{k,t}, \tag{1}$$

where the relative-importance weights are computed by the Bureau of Labor Statistics as the share of aggregate consumer expenditure devoted to each commodity category, with expenditures aggregated across households:

$$\omega_k = \frac{\sum_i e_{i,k}}{\sum_i E_i}, \quad E_i \equiv \sum_k e_{i,k}. \tag{2}$$

Each household contributes to the index weights in proportion to its total expenditure, so high-expenditure households have proportionally larger influence on the index than low-expenditure households. Define household i's personal inflation rate as the inflation rate experienced on its own consumption basket, $\pi_{i,t} \equiv \sum_k (e_{i,k}/E_i)\, p'_{k,t}$, where $p'_{k,t} = \log p_{k,t} - \log p_{k,t-1}$. The first result is a structural identity.

Lemma 1 (Aggregate–personal identity)**.** *Aggregate Consumer Price Index inflation equals the expenditure-share-weighted average of personal inflation rates:*

$$\pi_t^{CPI} \equiv \sum_k \omega_k p'_{k,t} = \sum_i \frac{E_i}{\sum_j E_j}\, \pi_{i,t}. \tag{3}$$

Proof. Substitute ω_k from (2) and switch the order of summation:

$$\sum_k \omega_k p'_{k,t} = \frac{1}{\sum_j E_j} \sum_i \sum_k e_{i,k} p'_{k,t} = \frac{1}{\sum_j E_j} \sum_i E_i \cdot \sum_k \frac{e_{i,k}}{E_i} p'_{k,t} = \sum_i \frac{E_i}{\sum_j E_j}\, \pi_{i,t}. \qquad \square$$

The identity is purely mechanical: it requires no behavioral assumption and states only that the macroeconomic object measured by the Bureau of Labor Statistics is, by construction, the expenditure-share-weighted average of personal inflation rates.

Households report a forecast of inflation in general, not of their own basket. Denote household i's reported forecast by $F_i \equiv E_i^t[\pi_{t+1}^{\text{CPI}}]$. To connect these responses to the identity (3), I make one assumption.

Assumption 1 (Own-basket forecasting)**.** *Each household forecasts aggregate inflation through its own price experience:* $F_i = E_i^t[\pi_{i,t+1}]$.

The assumption says that when asked about prices in general, a household reports the inflation it expects to face. It has strong support. Experiential-learning evidence shows that individuals form inflation expectations disproportionately from the inflation they have personally lived through [Malmendier & Nagel, 2016]; reported expectations track households' own grocery and shopping price exposure [D'Acunto et al., 2021]; and question-wording experiments find that "prices in general" responses load on personal price experience and coincide closely with what respondents report for the prices they themselves pay [Bruine de Bruin, van der Klaauw & Topa, 2011]. Realized personal inflation rates also differ systematically across the income distribution [Kaplan & Schulhofer-Wohl, 2017], so an own-basket forecast inherits a cross-sectional slope. Section 5 provides further support: the monotone income slope in reported expectations is the signature of own-basket forecasting, since a population all forecasting the same aggregate and differing only by noise would show no systematic relationship between income and reported inflation. Appendix B shows that the central result survives, as a lower bound, when the assumption is relaxed.

Under Assumption 1 the reported forecasts inherit the structure of personal inflation rates, and two aggregates of the *same* responses present themselves. Working with a unit mass of households, let $\bar{E} \equiv \int_0^1 E_j\, dj$ denote mean expenditure and let $s_i \equiv E_i/\bar{E}$ be the household's *consumption (expenditure) share*, normalized so that $\int_0^1 s_i\, di = 1$ and the average household has $s_i = 1$. The empirical literature uses the equal-weighted *average*, which assigns every household the same weight,

$$\bar{\pi}_t^{\text{avg}} \equiv \int_0^1 F_i\, di. \tag{4}$$

The structurally consistent object is the consumption-share-weighted *aggregate*,

$$\bar{\pi}_t^{\text{agg}} \equiv \int_0^1 s_i F_i\, di. \tag{5}$$

The aggregate is the household sector's forecast of the published index: by Assumption 1 and the identity (3), $\int_0^1 s_i F_i\, di = \int_0^1 s_i E_i^t[\pi_{i,t+1}]\, di$ combines households' own-inflation forecasts with the very expenditure weights the Bureau of Labor Statistics uses to construct the Consumer Price Index. The equal-weighted average instead forecasts an index in which every household basket counts the same, an index no one publishes.

Proposition 1 (Average versus aggregate)**.** *The average and aggregate expectations differ by the cross-sectional covariance between the household's consumption share and its reported inflation expectation:*

$$\bar{\pi}_t^{avg} - \bar{\pi}_t^{agg} = -\operatorname{Cov}_i\big(s_i, F_i\big). \tag{6}$$

The proof is a short covariance identity and is given in Appendix A.

Two points deserve emphasis. First, the two measures are computed from identical inputs, the same reported responses F_i, and differ only in the weights, so the comparison is not about whether survey responses can be trusted, which both take as given, but only about which weights are applied. Equation (6) shows the entire difference is the covariance between the consumption share and the reported belief; because higher-expenditure households report systematically lower inflation expectations (Section 5), this covariance is negative and the conventional average lies above the aggregate. Second, nothing here asserts that either measure is accurate. The aggregate expectation is a forecast assembled from households' beliefs; whether it coincides with realized inflation, whether households are rational, is tested in Section 6.4, not assumed.

3. A HANK framework with heterogeneous inflation beliefs

The accounting aggregator of Section 2 is the consumption-weighted combination of beliefs consistent with the published Consumer Price Index. This raises a question that the accounting cannot answer: is the consumption weight also the weight that matters for how household inflation expectations propagate through the macroeconomy? To answer it I take an otherwise standard heterogeneous-agent New Keynesian model and change one thing only: I let households hold heterogeneous subjective beliefs about next-period inflation, $E_i^t[\pi_{t+1}]$, which are precisely the objects the surveys measure. Everything else is the textbook environment [Galí, 2008]: constant-relative-risk-aversion preferences, a New Keynesian goods block, Erceg–Henderson–Levin sticky wages, and a nominal bond. No other belief is distorted and no other deviation from rational expectations is imposed, so the weights that emerge are not artifacts of an exotic friction but properties of the standard model read through belief heterogeneity. The household side carries no borrowing or short-sale constraints and a single traded bond, so the three aggregators I derive are *frictionless benchmarks*: they identify the belief that each equilibrium relation would weight in the absence of portfolio frictions, and Section 3.5 notes how the bond-pricing aggregator changes once such frictions bind. Taking households' inflation forecasts directly from surveys, rather than deriving them from rational expectations about the equilibrium, relates to Moll [2025], who argues that rational expectations about equilibrium prices are difficult to sustain in heterogeneous-agent models. The question pursued here is nonetheless distinct, and logically prior to how any single agent forms a price expectation: given a cross-section of such expectations, how should they be aggregated into the one number that enters each macroeconomic equilibrium condition.

The model's central message is a coincidence. The consumption-share aggregator that the accounting picks out in Section 2 is *exactly* the aggregate of beliefs that governs aggregate demand through the household Euler equation (Proposition 2): the weight that makes the survey consistent with the published index is the weight that drives household spending. Because the overwhelming reason economists and central banks track household inflation expectations is their effect on demand, this is the result that matters most, and it is the paper's main theoretical contribution. The model also clarifies what the consumption aggregator is *not*: two other equilibrium relations load on two other aggregates: the wage Phillips curve on the labor-income-share-weighted aggregate, and the pricing of nominal bonds on the wealth-share-weighted aggregate. These are the right objects for questions about wage dynamics or bond markets, but they are secondary to the demand channel. The one object that corresponds to none of these relations is the equal-weighted average the empirical literature uses.

3.1. Environment

Time is discrete and infinite, $t = 0, 1, 2, \ldots$. A unit continuum of households is indexed by $i \in [0, 1]$. Household i holds subjective beliefs represented by the conditional-expectation operator $E_i^t[\cdot]$; beliefs are heterogeneous across households and are *not* assumed to be rational or mutually consistent. These operators are the empirical objects of the survey: $E_i^t[\pi_{t+1}]$ is what household i reports. No restriction is placed on how they relate to the true data-generating process, so nothing in this section assumes households are right.

Preferences and goods. Each household consumes a constant-elasticity basket of differentiated goods with price index P_t; gross inflation is $\Pi_t \equiv P_t/P_{t-1}$ and $\pi_t \equiv \log \Pi_t$. (Allowing the basket to differ across households reproduces the heterogeneous personal inflation rates of Section 2; for the macro block I take a common basket, which is immaterial to the weights derived below.) Household i ranks streams by

$$E_i^0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_{i,t}^{1-\sigma}}{1-\sigma} - \chi \frac{L_{i,t}^{1+\varphi}}{1+\varphi} \right], \tag{7}$$

with discount factor $\beta \in (0, 1)$, inverse intertemporal elasticity of substitution $\sigma > 0$, inverse Frisch elasticity $\varphi > 0$, and labor-disutility weight $\chi > 0$.

The nominal bond. There is a single one-period nominal bond in fixed net supply $\bar{B}$ (government debt). A unit purchased at t costs Q_t dollars and pays one dollar at $t+1$, so its gross nominal return is $1 + i_t \equiv 1/Q_t$. Because the payoff is fixed in dollars, its *real* return between t and $t+1$ is

$$\frac{1+i_t}{\Pi_{t+1}} = (1+i_t)\frac{P_t}{P_{t+1}}, \tag{8}$$

which is eroded one-for-one by realized inflation. This inflation exposure is what makes the bond's equilibrium price depend on inflation expectations, and is central to Section 3.5.

Labor market. Household i is the monopolistic supplier of labor variety i and sets its nominal wage $W_{i,t}$ subject to Calvo rigidity (Section 3.4). A competitive labor packer combines the varieties into aggregate labor $L_t = \left(\int_0^1 L_{i,t}^{(\varepsilon_w-1)/\varepsilon_w} di\right)^{\varepsilon_w/(\varepsilon_w-1)}$, with elasticity of substitution $\varepsilon_w > 1$ and wage index $W_t = \left(\int_0^1 W_{i,t}^{1-\varepsilon_w} di\right)^{1/(1-\varepsilon_w)}$, and sells aggregate labor to firms. Cost minimization by the packer gives the labor-demand schedule each household faces, $L_{i,t} = (W_{i,t}/W_t)^{-\varepsilon_w} L_t$.

Firms and price-setting. The goods side is the textbook New Keynesian block. A competitive final-good producer bundles a continuum of intermediate goods $j \in [0,1]$ into output $Y_t = \left(\int_0^1 Y_{j,t}^{(\varepsilon_p-1)/\varepsilon_p} dj\right)^{\varepsilon_p/(\varepsilon_p-1)}$, with elasticity $\varepsilon_p > 1$, generating the demand curve $Y_{j,t} = (P_{j,t}/P_t)^{-\varepsilon_p} Y_t$ and the price index $P_t = \left(\int_0^1 P_{j,t}^{1-\varepsilon_p} dj\right)^{1/(1-\varepsilon_p)}$ that defines the inflation rate π_t households forecast. Each intermediate good j is produced by a monopolistically competitive firm using labor alone, $Y_{j,t} = A_t N_{j,t}$ with productivity A_t, so all firms share the nominal marginal cost $MC_t = W_t/A_t$. Prices are sticky à la Calvo: with probability $1-\theta_p$ a firm may reset its price at t, and otherwise keeps the previous one, so a price set at t still applies s periods later with probability θ_p^s. A firm that resets chooses $P_{j,t}^*$ to maximize the expected present value of real profits over the states in which the price remains in force,

$$\max_{P^*} E_t \sum_{s=0}^{\infty} \theta_p^s \Lambda_{t,t+s} \frac{P^* - MC_{t+s}}{P_{t+s}} Y_{j,t+s|t}, \qquad Y_{j,t+s|t} = \left(\frac{P^*}{P_{t+s}}\right)^{-\varepsilon_p} Y_{t+s}, \tag{9}$$

where $\Lambda_{t,t+s} = \beta^s (C_{t+s}/C_t)^{-\sigma}$ is the stochastic discount factor of the firm's owners and profits are rebated to households as dividends $D_{i,t}$. The first-order condition aggregates in the usual way to the price Phillips curve $\pi_t = \beta E_t[\pi_{t+1}] + \kappa_p \widehat{mc}_t$, with $\kappa_p = (1-\theta_p)(1-\beta\theta_p)/\theta_p$. Because the paper's question is how *household* inflation beliefs propagate, I hold the firm block at the common-expectations benchmark and inject belief heterogeneity only into households, so E_t in (9) is the common operator. Allowing firm-side heterogeneity would add a parallel price aggregator but leave the three household aggregators below unchanged.

Monetary and fiscal policy. The monetary authority sets the nominal rate on the bond through a Taylor rule, $\hat{\imath}_t = \phi_\pi \pi_t + \phi_y y_t + \nu_t$, with $\phi_\pi > 1$, output gap $y_t \equiv \log(Y_t/\bar{Y})$, and policy shock ν_t. The government issues the nominal bond in fixed supply $\bar{B}$ and runs a budget balanced by lump-sum taxes, which Ricardian households internalize. These blocks close the model; they play no role in the aggregation results, which concern only which households' beliefs enter each equilibrium condition. The household side, to which I now turn, is where those beliefs live.

3.2. The household's problem and its first-order conditions

Household i chooses consumption $C_{i,t}$, bond holdings $B_{i,t}$, and (when permitted) its reset wage, subject to the nominal flow budget constraint

$$P_t C_{i,t} + Q_t B_{i,t} = B_{i,t-1} + W_{i,t} L_{i,t} + D_{i,t} - T_{i,t}, \tag{10}$$

where $B_{i,t-1}$ is the maturing payoff of bonds bought last period, $W_{i,t} L_{i,t}$ is labor income, and $T_{i,t}$ are lump-sum taxes. Let $\lambda_{i,t}$ be the multiplier on (10).

Consumption and bond optimality. The first-order conditions for $C_{i,t}$ and $B_{i,t}$ are

$$C_{i,t}^{-\sigma} = \lambda_{i,t} P_t, \qquad \lambda_{i,t} Q_t = \beta E_i^t[\lambda_{i,t+1}]. \tag{11}$$

Eliminating $\lambda_{i,t} = C_{i,t}^{-\sigma}/P_t$ and using $Q_t = 1/(1+i_t)$ gives the household Euler equation for the inflation-exposed bond,

$$C_{i,t}^{-\sigma} = \beta(1+i_t)\,E_i^t\left[\frac{C_{i,t+1}^{-\sigma}}{\Pi_{t+1}}\right], \tag{12}$$

in which the stochastic discount factor is deflated by realized inflation Π_{t+1}, per (8).

Log-linearization. Let a zero-inflation steady state have $\bar{\Pi} = 1$ and $\beta(1+\bar{\imath}) = 1$, and write log-deviations as $c_{i,t} \equiv \log(C_{i,t}/\bar{C}_i)$, $\hat{\imath}_t \equiv \log\big((1+i_t)/(1+\bar{\imath})\big)$, and $\pi_{t+1} \equiv \log \Pi_{t+1}$. Taking logs of (12) and linearizing the expectation term to first order around the steady state,

$$-\sigma c_{i,t} = \hat{\imath}_t + E_i^t[-\sigma c_{i,t+1} - \pi_{t+1}], \tag{13}$$

which rearranges to the individual Euler equation

$$c_{i,t} = E_i^t[c_{i,t+1}] - \frac{1}{\sigma}\big(\hat{\imath}_t - E_i^t[\pi_{t+1}]\big). \tag{14}$$

Household i substitutes consumption across periods in response to the real rate $\hat{\imath}_t - E_i^t[\pi_{t+1}]$ it perceives. Whether spending responds to inflation expectations with this sign is debated empirically [Bachmann, Berg & Sims, 2015]; what follows concerns *which* households' expectations enter each aggregate relation, independent of the sign of the individual response.

3.3. Aggregate demand: the consumption-share aggregator

Let $C_t \equiv \int_0^1 C_{i,t}\,di$ be aggregate consumption, $\bar{C} \equiv \int_0^1 \bar{C}_i\,di$ its steady-state level, and $\omega_i \equiv \bar{C}_i/\bar{C}$ household i's share of steady-state consumption, with $\int_0^1 \omega_i\,di = 1$. A first-order expansion of $C_t = \int_0^1 \bar{C}_i e^{c_{i,t}}\,di$ around the steady state gives $C_t \approx \bar{C}\big(1 + \int_0^1 \omega_i c_{i,t}\,di\big)$, so the aggregate consumption gap $c_t \equiv \log(C_t/\bar{C})$ is the consumption-share-weighted average of individual gaps:

$$c_t = \int_0^1 \omega_i\, c_{i,t}\,di. \tag{15}$$

Substituting the individual Euler equation (14) into (15) and using that the nominal rate $\hat{\imath}_t$ is common across households with $\int_0^1 \omega_i\,di = 1$,

$$c_t = \int_0^1 \omega_i\, E_i^t[c_{i,t+1}]\,di - \frac{1}{\sigma}\left(\hat{\imath}_t - \int_0^1 \omega_i\, E_i^t[\pi_{t+1}]\,di\right). \tag{16}$$

Defining the consumption-share-weighted aggregates

$$\bar{E}_t^C[c_{t+1}] \equiv \int_0^1 \omega_i\, E_i^t[c_{i,t+1}]\,di, \qquad \bar{E}_t^C[\pi_{t+1}] \equiv \int_0^1 \omega_i\, E_i^t[\pi_{t+1}]\,di, \tag{17}$$

equation (16) collapses to the aggregate Euler equation

$$c_t = \bar{E}_t^C[c_{t+1}] - \frac{1}{\sigma}\big(\hat{\imath}_t - \bar{E}_t^C[\pi_{t+1}]\big). \tag{18}$$

The inflation expectation that governs aggregate demand is the consumption-share-weighted aggregate $\bar{E}_t^C[\pi_{t+1}]$, not the equal-weighted mean. The next result connects this to the accounting aggregator of Section 2.

Proposition 2 (Demand aggregator equals the accounting aggregator)**.** *At the steady state, household i's consumption share equals its expenditure share: $\omega_i = \bar{C}_i/\bar{C} = \bar{E}_i/\bar{E} = s_i$, where $\bar{E}_i \equiv P\bar{C}_i$ is nominal expenditure. Hence the aggregate-demand aggregator coincides with the accounting aggregator of Proposition 1:*

$$\bar{E}_t^C[\pi_{t+1}] = \int_0^1 s_i E_i^t[\pi_{t+1}]\,di = \bar{\pi}_t^{\text{agg}}. \tag{19}$$

Proof. With a common price level P, nominal expenditure is $\bar{E}_i = P\bar{C}_i$, so $s_i = \bar{E}_i/\bar{E} = P\bar{C}_i/(P\bar{C}) = \bar{C}_i/\bar{C} = \omega_i$. Substituting into (17) and comparing with (5) gives the claim. □

Proposition 2 is the model's main reassurance: the weight that makes the survey consistent with the published index is exactly the weight that the structural model assigns to the aggregate-demand channel. Empirically this consumption-share aggregator is measured directly from the Consumer Expenditure Survey, whose nonparametric expenditure schedule has a cross-sectional consumption-income elasticity of about 0.45 [Aguiar & Bils, 2015, Fagereng, Holm & Natvik, 2021]; its magnitude is quantified in Section 6.

Borrowing constraints. The derivation puts every household on its Euler equation, the frictionless benchmark. The defining friction of heterogeneous-agent models is that some households are not: a fraction are hand-to-mouth, consuming their current income and not substituting intertemporally, so their consumption responds to income rather than to the expected real rate. This does not change how beliefs are aggregated for aggregate demand; it changes only whose beliefs count and how strongly. Splitting the aggregate consumption gap (15) into unconstrained households U and hand-to-mouth households and substituting the Euler equation (14) only for the former, the expected-inflation term in aggregate demand becomes

$$\frac{1}{\sigma}\,\Omega_U\,\bar{E}_t^{C,U}[\pi_{t+1}], \qquad \bar{E}_t^{C,U}[\pi_{t+1}] \equiv \frac{1}{\Omega_U}\int_U \omega_i E_i^t[\pi_{t+1}]\,di, \qquad \Omega_U \equiv \int_U \omega_i\,di, \tag{20}$$

where $\Omega_U < 1$ is the unconstrained share of aggregate consumption. Two things change relative to (18), and only the first is about strength. The sensitivity of demand to expected inflation is scaled down by Ω_U: aggregate demand responds less than the frictionless benchmark because the constrained households do not react, the standard muted-intertemporal-substitution result of heterogeneous-agent models. But the belief that enters is still a *consumption-share-weighted* aggregate, now taken over the unconstrained households and renormalized; the weighting rule is unchanged, and only the population shrinks to those who actually substitute. Because the consumption weight already assigns little mass to the low-consumption households who are disproportionately constrained, $\bar{E}_t^{C,U}$ stays close to the frictionless aggregate $\bar{E}_t^C$, so the friction dampens the strength of the expectations channel without altering the object on which it acts. This restriction to the unconstrained is also what the evidence calls for: the spending of higher-income, financially unconstrained households responds more strongly to inflation expectations than that of constrained households [Weber et al., 2022, Coibion et al., 2023], exactly the households the consumption-share weight already emphasizes.

3.4. *The wage Phillips curve: the labor-income-share aggregator*

It is worth pausing on who sets wages in this model, because the answer is what lets household inflation beliefs enter wage dynamics. I use the standard sticky-wage structure of Erceg, Henderson & Levin [2000], which divides nominal rigidity between the two sides of the economy: firms set prices, and *households* set wages. Labor is differentiated, so each household i is the monopolistic supplier of its own labor variety and posts the wage $W_{i,t}$ for it; a competitive labor packer then bundles these varieties into the aggregate labor input that firms hire, and firms take the resulting wage index as given. Wage stickiness therefore lives at the household level: the household, not the firm, is the agent who chooses the nominal wage, and it does so subject to Calvo rigidity. This is the standard textbook environment [Galí, 2008], but the assignment matters here for a specific reason. Because the household sets its own wage, its reset wage depends on *its own* expected inflation, through the future price level it anticipates; heterogeneous beliefs thus enter wage formation household by household, and aggregating across wage-setters is what produces the labor-income-share aggregator below. Were firms the wage-setters, there would be no individual-belief heterogeneity in wage-setting and no such aggregator.

Consider a household that draws a Calvo opportunity to reset its wage at t, which arrives with probability $1-\theta_w$ each period. With the complementary probability θ_w the chosen wage carries over unchanged, so a wage set at t is still in force s periods later with probability θ_w^s. Its wage choice follows from the household's lifetime utility (7), restricted to the terms the wage affects. Posting W^* obliges the household, in every future period $t+s$ in which the wage still applies, to supply whatever labor the demand schedule calls for at that wage,

$$L_{i,t+s|t} = \left(W^*/W_{t+s}\right)^{-\varepsilon_w} L_{t+s}, \tag{21}$$

where the notation $L_{i,t+s|t}$ records that these are the hours at $t+s$ under a wage last reset at t, and it earns nominal wage income $W^* L_{i,t+s|t}$. That income enters utility only by relaxing the period budget constraint, whose shadow value is the marginal utility of nominal income,

$$\lambda_{i,t+s} = \frac{C_{i,t+s}^{-\sigma}}{P_{t+s}}, \tag{22}$$

the multiplier on the budget constraint, equal to the marginal utility of consumption $C_{i,t+s}^{-\sigma}$ converted to dollars by the price level. The income is therefore worth $\lambda_{i,t+s} W^* L_{i,t+s|t}$ in utility, against which the household nets the disutility $\chi L_{i,t+s|t}^{1+\varphi}/(1+\varphi)$ of supplying the hours, taken from (7). Discounting by β^s, weighting each future date by the survival probability θ_w^s, and taking the household's subjective expectation E_i^t, the resetting household chooses W^* to maximize the net contribution of the wage to lifetime utility,

$$\max_{W^*} E_i^t \sum_{s=0}^{\infty} (\beta\theta_w)^s \left[\lambda_{i,t+s} W^* L_{i,t+s|t} - \chi \frac{L_{i,t+s|t}^{1+\varphi}}{1+\varphi} \right]. \tag{23}$$

Each term is the period payoff from the wage: the utility value of the income it brings in, minus the disutility of the hours it obliges the household to work. Differentiating (23) with respect to W^* and using $\partial L_{i,t+s|t}/\partial W^* = -\varepsilon_w L_{i,t+s|t}/W^*$ from (21), so that $\partial(W^* L_{i,t+s|t})/\partial W^* = (1-\varepsilon_w)L_{i,t+s|t}$, the first-order condition is

$$E_i^t \sum_{s=0}^{\infty} (\beta\theta_w)^s L_{i,t+s|t} \left[(1-\varepsilon_w)\lambda_{i,t+s} + \varepsilon_w \frac{\chi L_{i,t+s|t}^{\varphi}}{W^*} \right] = 0, \tag{24}$$

which rearranges to set the reset wage as a markup $\mu_w \equiv \varepsilon_w/(\varepsilon_w - 1)$ over a discounted average of the nominal marginal rate of substitution $\chi L_{i,t+s|t}^{\varphi}/\lambda_{i,t+s}$:

$$W_{i,t}^* = \mu_w \frac{E_i^t \sum_s (\beta\theta_w)^s L_{i,t+s|t}\, \chi L_{i,t+s|t}^{\varphi}}{E_i^t \sum_s (\beta\theta_w)^s L_{i,t+s|t}\, \lambda_{i,t+s}}. \tag{25}$$

Log-linearizing (25) around the steady state, the reset wage is the expected discounted path of the household's nominal marginal rate of substitution $mrs_{i,t+s} + p_{t+s}$ (with $mrs_{i,t+s} \equiv \varphi \ell_{i,t+s} + \sigma c_{i,t+s}$ the log real MRS and $p_{t+s} \equiv \log P_{t+s}$):

$$w_{i,t}^* = (1-\beta\theta_w) \sum_{s=0}^{\infty} (\beta\theta_w)^s E_i^t [mrs_{i,t+s} + p_{t+s}]. \tag{26}$$

The object that matters for inflation is not any single wage but the growth of aggregate nominal labor income $\Omega_t \equiv \int_0^1 W_{i,t} L_{i,t}\, di$, which is the wage bill that enters firms' marginal cost. Let $\eta_i \equiv \bar{W}_i \bar{L}_i/(\bar{W}\bar{L})$ be household i's steady-state share of the wage bill (its labor-income share), with $\int_0^1 \eta_i\, di = 1$. As in (15), a first-order expansion gives the aggregate wage-bill gap as the labor-income-share-weighted average of individual wage and hours deviations,

$$\omega_t^{\Omega} \equiv \log(\Omega_t/\bar{\Omega}) = \int_0^1 \eta_i (w_{i,t} + \ell_{i,t})\, di, \tag{27}$$

so that labor-cost inflation $\pi_t^{\Omega} \equiv \omega_t^{\Omega} - \omega_{t-1}^{\Omega}$ inherits the labor-income weights. Because each resetting household's wage (26) loads on its own expected inflation through the price-level path $p_{t+s} = p_t + \sum_{u=1}^{s} \pi_{t+u}$, aggregating the reset-wage equations into (27) with weights η_i delivers a wage Phillips curve whose expectation term is the labor-income-share-weighted aggregate. Collecting terms in the standard way [Erceg, Henderson & Levin, 2000, Galí, 2008],

$$\pi_t^{\Omega} = \beta\,\bar{E}_t^{W}[\pi_{t+1}^{\Omega}] + \kappa_w\big(\overline{mrs}_t - (w_t - p_t)\big), \qquad \bar{E}_t^{W}[\pi_{t+1}] \equiv \int_0^1 \eta_i\, E_i^t[\pi_{t+1}]\, di, \tag{28}$$

with slope $\kappa_w = (1-\theta_w)(1-\beta\theta_w)/\big(\theta_w(1+\varepsilon_w\varphi)\big)$ and $\overline{mrs}_t$ the labor-income-weighted average MRS. The inflation expectation that governs wage and labor-cost dynamics is the *labor-income-share-weighted* aggregate $\bar{E}_t^{W}[\pi_{t+1}]$: high-wage-bill households contribute proportionally more to aggregate labor income and so to aggregate wage inflation, exactly as high-consumption households dominate (18). Because labor income scales roughly linearly with total income for wage earners, I weight by income; its magnitude is quantified in Section 6.

3.5. Bond pricing: the wealth-share aggregator

The third channel is the pricing of the inflation-exposed bond. Log-linearizing the household's bond Euler equation (14) states that every household holding the bond perceives the same after-inflation real return and equates it to its own expected consumption growth:

$$\hat{\imath}_t - E_i^t[\pi_{t+1}] = \sigma\, E_i^t[\Delta c_{i,t+1}], \qquad \Delta c_{i,t+1} \equiv c_{i,t+1} - c_{i,t}. \tag{29}$$

The nominal rate $\hat{\imath}_t$ is common, but the expected inflation and consumption-growth terms are household-specific, so beliefs must be reconciled by the market. The reconciliation is through market clearing for the bond, and it weights households by the size of their positions.

Let $A_{i,t} \equiv B_{i,t}/P_t$ be household i's real bond holdings, $\bar{A}_i$ its steady-state level, $\bar{A} \equiv \int_0^1 \bar{A}_i\, di = \bar{B}/P$ aggregate real bond supply, and $\zeta_i \equiv \bar{A}_i/\bar{A}$ household i's share of bond holdings, its wealth share, with $\int_0^1 \zeta_i\, di = 1$. Two equivalent routes give the equilibrium relation.

Route 1: market clearing of bond demand. Bond demand comes straight from the household's first-order condition for bonds, which is the Euler equation (14). Read as a saving decision, it says that, holding expected future consumption fixed, a household perceiving a higher real return $\hat{\imath}_t - E_i^t[\pi_{t+1}]$ lowers current consumption by $1/\sigma$ per unit of that return: $\partial c_{i,t}/\partial\big(\hat{\imath}_t - E_i^t[\pi_{t+1}]\big) = -1/\sigma$. The flow budget constraint (10) turns the foregone consumption into saving, and the bond is the only asset, so the consumption a household gives up it carries into bonds. A higher perceived real return therefore raises real bond demand, with a strength governed by the elasticity of intertemporal substitution $1/\sigma$. Because preferences are homothetic, this response is the same fraction of holdings for every household, so in absolute terms each scales its position with its steady-state holdings $\bar{A}_i$:

$$A_{i,t} = \bar{A}_i\Big[\psi\big(\hat{\imath}_t - E_i^t[\pi_{t+1}]\big) + \xi_{i,t}\Big], \tag{30}$$

where the rate response $\psi > 0$ is set by $1/\sigma$ and is identical across households because they share σ, and $\xi_{i,t}$ collects the income and initial-wealth terms that do not depend on the rate. This common per-dollar response is what makes wealth the weight: a household with twice the holdings moves the market twice as much for the same belief. Bond-market clearing requires aggregate demand to equal fixed supply, $\int_0^1 A_{i,t}\,di = \bar{A}\,a_t$ for an exogenous supply path a_t. Dividing by $\bar{A}$ and using $\zeta_i = \bar{A}_i/\bar{A}$,

$$a_t = \psi\Big(\hat{\imath}_t - \int_0^1 \zeta_i E_i^t[\pi_{t+1}]\,di\Big) + \int_0^1 \zeta_i\,\xi_{i,t}\,di. \tag{31}$$

Solving for the nominal rate that clears the market,

$$\hat{\imath}_t = \int_0^1 \zeta_i E_i^t[\pi_{t+1}]\,di + \frac{1}{\psi}\Big(a_t - \int_0^1 \zeta_i\,\xi_{i,t}\,di\Big). \tag{32}$$

Route 2: the wealth-weighted Euler equation. Equivalently, multiply each household's bond Euler (29) by its wealth share ζ_i and integrate. This is the aggregation that respects market clearing, because each household's marginal contribution to aggregate bond demand scales with its position:

$$\hat{\imath}_t = \int_0^1 \zeta_i E_i^t[\pi_{t+1}]\,di + \sigma\int_0^1 \zeta_i E_i^t[\Delta c_{i,t+1}]\,di. \tag{33}$$

Either route yields the equilibrium Fisher equation

$$\hat{\imath}_t = \bar{r}_t + \bar{E}_t^A[\pi_{t+1}], \qquad \bar{E}_t^A[\pi_{t+1}] \equiv \int_0^1 \zeta_i E_i^t[\pi_{t+1}]\,di, \tag{34}$$

where $\bar{r}_t$ is the equilibrium real rate (the wealth-weighted expected consumption-growth term in Route 2, or the supply term in Route 1). The inflation expectation embedded in the equilibrium bond price is the *wealth-share-weighted* aggregate $\bar{E}_t^A[\pi_{t+1}]$, the logic by which heterogeneous-belief asset-pricing models let the wealthier and more optimistic agents move prices the most [Adam, Marcet & Nagel, 2016].

This wealth-weighted aggregate is a *frictionless benchmark*. It follows from the linearized bond Euler under common preferences and *interior* holdings, with every household holding the bond, so each contributes to the price in proportion to its wealth. With portfolio frictions, the object is different: Reis [2021], in his account of the discrepancy between survey and market inflation expectations, shows that once short-sale and borrowing constraints bind, traders take corner positions and the bond is priced by the belief of the *marginal* investor, whose expectation can lie far in the tail of the wealth distribution rather than at its weighted mean. The wealth-share-weighted aggregate is the unconstrained limit of that marginal-investor belief, and the right "market" counterpart to the consumption- and labor-income-share aggregators on the household side; mapping it to observed market-implied expectations, where these frictions are first order, is left to future work.

Proposition 3 (Three object-dependent aggregators)**.** *In the model of Section 3.1, the subjective inflation* *expectation that enters the equilibrium relation is*

$$
\begin{aligned}
&\textit{aggregate demand (18):} && \bar{E}_t^C[\pi_{t+1}] = \int_0^1 \omega_i E_i^t[\pi_{t+1}]\,di, && \omega_i = \bar{C}_i/\bar{C}\ (= s_i),\\
&\textit{wage Phillips curve (28):} && \bar{E}_t^W[\pi_{t+1}] = \int_0^1 \eta_i E_i^t[\pi_{t+1}]\,di, && \eta_i = \bar{W}_i\bar{L}_i/(\bar{W}\bar{L}),\\
&\textit{bond pricing (34):} && \bar{E}_t^A[\pi_{t+1}] = \int_0^1 \zeta_i E_i^t[\pi_{t+1}]\,di, && \zeta_i = \bar{A}_i/\bar{A}.
\end{aligned}
$$

Each weight integrates to one, so each aggregator is a weighted mean of the same survey responses; the equal-weighted average $\int_0^1 E_i^t[\pi_{t+1}]\,di$ corresponds to none of the three unless the weights are degenerate (all households identical).

The three weights are ordered by how concentrated the underlying stake is across the income distribution. Consumption is the least concentrated (the consumption-income elasticity is well below one), labor income is intermediate (roughly proportional to income), and wealth is the most concentrated of all. Because the high-stake households are systematically the low-expectation households, the more concentrated the weight, the larger the downward correction relative to the equal-weighted average. Section 6.2 confirms this ordering in the data once the surveys and the Consumer Expenditure Survey are in hand.

The role of the model is not to privilege one channel but to make a single general point: *the correct weight on an agent's belief is that agent's stake in the object the belief is being used to forecast.* Whenever a heterogeneous-belief population is aggregated into an equilibrium relation, the inflation expectation that enters is the belief-weighted average in which each agent's weight is its share of the aggregate quantity that relation governs. The consumption-, labor-income-, and wealth-share aggregators are three instances of this principle; the equal-weighted average is the special case that assigns every agent the same stake, which no equilibrium relation does. Proposition 2 singles out the consumption-share aggregator because aggregate demand is the dominant reason to track household inflation expectations, but the principle itself is channel-independent.

I state the principle for households because household survey beliefs are the object I aggregate; the analysis is deliberately silent about firms, whose inflation expectations are elicited in separate surveys. The same logic, however, extends directly to the supply side. If firms hold heterogeneous beliefs about future inflation, aggregating their forward-looking Calvo price-setting decisions into the New Keynesian price Phillips curve weights each firm's belief by its share of the aggregate it determines (output, equivalently sales or value added), exactly as the wage Phillips curve weights households by their labor-income share. The object that governs aggregate supply is therefore the *sales-share-weighted* aggregate of firm inflation expectations, and the equal-weighted average of a firm survey would mismeasure it for the same reason it mismeasures the household objects here. Constructing that firm-side aggregate from firm-level survey data is the natural counterpart to the present exercise and is left to future work.

4. Data

The empirical analysis combines microdata from the Michigan Survey of Consumers, the New York Fed Survey of Consumer Expectations and its Household Finance Survey supplement, professional forecaster data from the Survey of Professional Forecasters, and realized macroeconomic series from FRED.

The Michigan Survey of Consumers public-use microdata span January 1978 through January 2025 with 326,940 respondent-month observations across 565 monthly survey waves. The principal inflation variable is `PX1`, the open-ended point forecast of one-year-ahead inflation in percent; I retain numeric responses in the range $[-10, 50]$ percent and treat values 95 and above as missing. Household pre-tax annual income is reported as a continuous variable `INCOME`, top-coded at $5,000,000. The sampling weight `WT` adjusts for population representativeness by age and education.

The Survey of Consumer Expectations is a monthly internet survey of approximately 1,300 household heads launched in June 2013 with a rotating panel structure. I use the public microdata covering June 2013 through February 2019, with 89,665 respondent-month observations across approximately 13,031 unique respondents. The principal expectation variable is `Q9_mean`, the density-implied subjective mean of one-year-ahead inflation derived by fitting a generalized beta distribution to ten reported probability bins. The survey also records an 11-bracket household-income variable `Q47`, asked on the respondent's first wave and forward-filled within respondent by `userid` (88,716 observations with valid `Q47`); I use it only for the bracket-level description of the income slope in Table 1. All belief weighting in the Survey of Consumer Expectations instead uses the continuous income measure from the Household Finance Survey supplement, described next.

The Household Finance Survey is a quarterly supplement to the Survey of Consumer Expectations with 6,809 unique respondents. The continuous income variable `b1_1` reports total annual household income in dollars; I exclude observations below $1,000 or above $5,000,000 as outliers. The supplement also reports total household financial wealth (the value of savings and investments, `d16new_1`). Merging it with the core inflation panel by `userid` yields 42,357 respondent-month observations with valid expectations, continuous income, and wealth; this is the sample on which all the Survey of Consumer Expectations aggregators (consumption, labor-income, and wealth) are computed.

Household expenditure is observed in neither expectations survey, so I construct each respondent's consumption share from the Consumer Expenditure Survey and merge it in by income. The construction has two steps: estimating a nonparametric consumption-share schedule in the Consumer Expenditure Survey, and mapping that schedule onto each survey respondent through her position in the income distribution.

Constructing the schedule. I pool the Consumer Expenditure Survey Interview consumer-unit files for 2013–2024, roughly 250,000 consumer-unit-quarters. For each consumer unit I take total quarterly expenditure (`TOTEXPPQ`) multiplied by four to annualize, pre-tax income (`FINCBTXM`, the Bureau's mean of the five multiply-imputed income values), and the final population weight (`FINLWT21`) divided by four so that the four pooled quarters represent the population once; I drop consumer units with income below $500 or non-positive expenditure or weight. I then sort consumer units by income, partition them into 200 population-weighted income-percentile bins, and compute the weight-mean annual expenditure within each bin. Dividing each bin's mean expenditure by population-mean expenditure yields a consumption-share schedule $s(p)$ on the income percentile $p \in (0,1)$, normalized so that the population-weighted mean of s is one. This schedule is simply the binned conditional mean $E[\text{expenditure} \mid \text{income rank}]$ expressed as a share; it imposes no functional form and in particular no constant-elasticity (log-linear) restriction, though the implied expenditure–income elasticity is close to 0.45 throughout, in the range of micro estimates in the literature [Aguiar & Bils, 2015, Fagereng, Holm & Natvik, 2021] and well below one, so that expenditure is far more compressed across the distribution than income. The schedule is increasing and concave. Between bin centers I interpolate $s(p)$ linearly; below the lowest and above the highest bin center I hold it flat, so no respondent is assigned a value extrapolated beyond the observed Consumer Expenditure Survey support.

Mapping the schedule to the surveys. Because no household appears in both an expectations survey and the Consumer Expenditure Survey, I link them through income. For each survey respondent I compute her percentile p in the income distribution of her own survey, sorting by reported income (`INCOME` in Michigan, $b_{1,1}$ in the Survey of Consumer Expectations) and accumulating the survey sampling weights; the percentile is computed within survey-year in the Michigan panel, where nominal incomes trend over four decades, and over the pooled 2013–2019 window in the Survey of Consumer Expectations. I then read her consumption share off the schedule, $s_i = s(p_i)$, and renormalize the assigned shares to mean one within each survey sample. Working in income rank rather than nominal dollars is what lets a single schedule apply to the entire Michigan sample, including the decades before the Consumer Expenditure Survey microdata begin, and is justified by the stability of the rank–expenditure relationship, whose elasticity is about 0.45 in every year. As a check on this rank-based, pooled mapping, I also re-estimate the schedule year by year over 2013–2024 and match in same-year nominal dollars; the resulting wedges are nearly identical. The procedure assigns the average expenditure of Consumer Expenditure Survey households at a given income rank to expectations-survey households at the same rank, and so identifies the consumption share under the assumption that, conditional on income, a household's expenditure is independent of its inflation belief.

The professional benchmark for inflation is the median forecast of annual-average Consumer Price Index inflation for the next calendar year (`CPIB`) from the Survey of Professional Forecasters. Realized inflation is computed as $100 \cdot (\text{CPIAUCSL}_{t+12}/\text{CPIAUCSL}_t - 1)$.

For each survey month t I compute the equal-weighted aggregate and the structurally consistent alternatives as $\bar{X}_t^S = \sum_i w_i S_i X_i / \sum_i w_i S_i$, where w_i is the survey sampling weight, S_i is the household-level economic stake (with $S_i \equiv 1$ for the equal-weighted aggregate), and X_i is the belief variable. The stakes are the consumption share (the Consumer Expenditure Survey expenditure schedule evaluated at the household's income rank), income (the labor-income stake), and, in the Survey of Consumer Expectations only, household financial wealth.

5. Inflation expectations decline with every economic stake

Each of the three structural aggregators differs from the equal-weighted average by the cross-sectional covariance between its weight and the reported expectation (Proposition 1), so all three corrections work in the same direction if and only if expected inflation falls with each of the three stakes. The cleanest starting point is income, which is observed directly. Table 1 reports one-year expectations across the eleven income brackets of the Survey of Consumer Expectations: the mean falls almost monotonically from 4.72 percent in the \$10–20k bracket to 2.80 percent above \$200k, and the decline is present at the median as well, so it is not driven by a few respondents reporting very high values. The same income slope appears over the full Michigan sample across within-year income terciles (5.15, 4.54, 4.11 percent), so it is not specific to the recent period, and it holds in both surveys despite their different question formats.

Table 1: Inflation expectations by income bracket, Survey of Consumer Expectations

`Q47` bracket	Income range	n	Mean (%)	Median (%)
1	Less than \$10k	2,779	4.21	2.78
2	\$10–20k	5,824	4.72	3.07
3	\$20–30k	7,420	4.42	3.00
4	\$30–40k	7,330	4.52	3.00
5	\$40–50k	7,696	3.89	3.00
6	\$50–60k	7,913	4.09	3.00
7	\$60–75k	10,807	3.70	2.97
8	\$75–100k	12,346	3.50	2.88
9	\$100–150k	13,509	3.25	2.80
10	\$150–200k	5,777	3.22	2.78
11	More than \$200k	4,989	2.80	2.58

The aggregators weight by consumption, labor income, and wealth, however, not by income itself, so what matters is whether the decline survives when households are ranked by each of these stakes. Figure 1 shows that it does. Sorting Survey of Consumer Expectations households into quintiles of the consumption share, the labor-income share, and the wealth share, mean expectations fall across all three, from about 4.2 percent in the lowest-stake quintile to about 3.0 percent in the highest. Because the consumption and labor-income stakes both rise with income they rank households the same way, so those two panels coincide; wealth re-ranks households on an independent dimension, and the decline survives there too, which rules out the income ranking driving the result. The panels differ in how concentrated the stake is, which is what will order the corrections: the top decile holds about 2.4 times the average consumption, 3.4 times the average labor income, and 6.8 times the average wealth.

The slope is not merely statistical; it reflects real differences in inflation exposure. Lower-income households allocate more of their spending to food, energy, and rent, whose prices grew faster than the overall

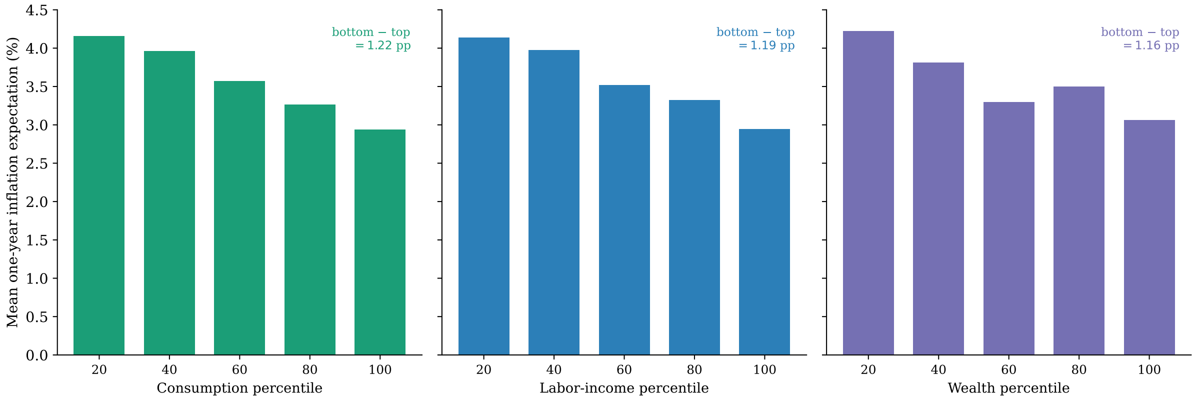

Figure 1: Mean one-year inflation expectation by stake percentile, Survey of Consumer Expectations 2013–2019. Households are sorted into population-weighted quintiles of the consumption share (the Consumer Expenditure Survey expenditure schedule evaluated at the household's income rank), the labor-income share (income), and the wealth share (savings and investments, winsorized at the 99th percentile); the horizontal axis is the percentile of the relevant stake and each bar is the weighted-mean expectation in that quintile. Expectations fall with every stake. The consumption and labor-income panels rank households by income and so are nearly identical, differing only in stake concentration (the top decile holds about 2.4, 3.4, and 6.8 times the average consumption, labor income, and wealth, respectively); wealth is an independent sort.

index [D'Acunto et al., 2021]; personal Consumer Price Index inflation is genuinely higher for low-income households, often by a full percentage point [Kaplan & Schulhofer-Wohl, 2017]; and recent posted-price inflation has been higher in grocery than in service categories [Cavallo, Lippi & Miyahara, 2024]. The slope in expectations is therefore at least partly a reflection of the inflation households actually face rather than belief distortion.

6. Averaging versus aggregating

6.1. The aggregate versus the average

How much can the choice of weight matter? Table 2 reports the share of each aggregation weight held by three income tiers. Under equal weighting a tier's weight is just its population share, roughly a third each. The structural weights redistribute sharply toward the top: the above-$100k tier holds 32 percent of the population but 53 percent of the consumption weight, 63 percent of the labor-income weight, and 58 percent of the wealth weight, while the below-$50k tier falls from a third of the population to between 10 and 17 percent of the structural weights. Because the lower tiers are the more pessimistic, this redistribution is what makes every structural aggregate lie below the equal-weighted average, and the more concentrated the stake the larger the shift.

Figure 2 plots the household sector's one-year inflation expectation under each weighting against the Survey of Professional Forecasters median. Panel A uses the Michigan survey, which is long and large but collects no balance-sheet data. The consumption-weighted aggregate, the object that the accounting picks out and that governs aggregate demand (Proposition 2), lies below the equal-weighted average in

Table 2: Weight share by income tier under each aggregation scheme, Survey of Consumer Expectations

Income range	Equal weight	Consumption weight	Labor-income weight	Wealth weight
Below \$50k	33.2%	16.7%	10.3%	13.1%
\$50k to \$100k	35.2%	30.5%	26.8%	29.2%
Above \$100k	31.6%	52.8%	63.0%	57.7%

Notes. Each column is the share of the corresponding aggregation weight held by the income tier. Equal weight equals the population (sampling-weighted) share. Computed on the Survey of Consumer Expectations respondents with observed financial wealth ($n = 42{,}357$), the sample on which the survey's aggregators are built; this subsample is somewhat higher-income than the full survey.

essentially every month since 1982; the labor-income-weighted aggregate lies lower still. The shaded band between the average and the consumption aggregate is the aggregation correction, and it is exactly the part of the household–professional gap attributable to weights rather than to beliefs. The aggregators move with the average but sit closer to the professional forecast throughout, and the band widens most in the post-2020 surge. Panel B adds the wealth-weighted aggregate over the Survey of Consumer Expectations window, the one survey with household balance sheets; the wealth aggregate lies below all the others, consistent with wealth being the most concentrated stake.

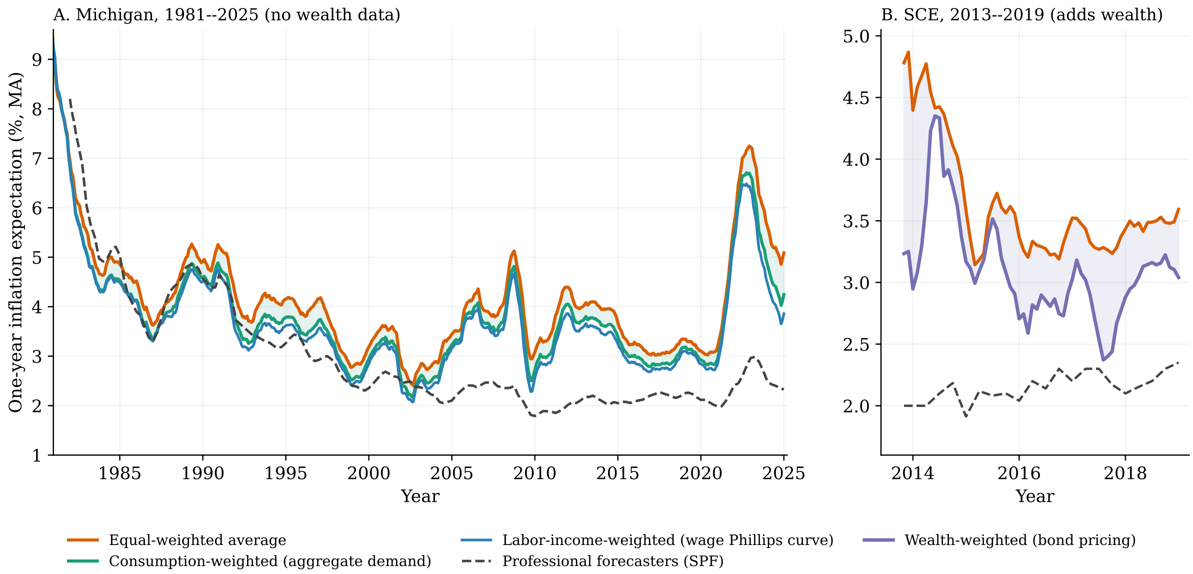

Figure 2: Household one-year inflation expectation by aggregator. *Panel A*: Michigan Survey of Consumers, 1981–2025, twelve-month moving averages, showing the equal-weighted average and the consumption- and labor-income-weighted aggregates (Michigan collects no wealth data); the shaded band is the gap between the average and the consumption aggregate. *Panel B*: Survey of Consumer Expectations, 2013–2019, three-month moving averages, which adds the wealth-weighted aggregate (shaded band: average to wealth aggregate). The equal-weighted average uses the survey sampling weight; the structural aggregates additionally weight by the Consumer Expenditure Survey consumption share, by income, and by household financial wealth. SPF is the Survey of Professional Forecasters median.

6.2. *The aggregate is significantly less pessimistic*

The gap is large and statistically decisive. Table 3 reports each aggregator's wedge relative to the equal-weighted average, with standard errors from a bootstrap that resamples survey months for Michigan and households for the Survey of Consumer Expectations. In the Michigan sample the consumption aggregate is 0.32 percentage points below the average ($t = -31$) and the labor-income aggregate 0.87 points below ($t = -15$); both reject equality with the average at any conventional level. The Survey of Consumer Expectations reproduces the consumption and labor-income wedges on an independent survey and adds the wealth aggregator, which Michigan cannot measure: weighting by household financial wealth lowers the aggregate by 0.51 percentage points ($t = -12$). All three structural aggregators are significantly less pessimistic than the equal-weighted average, and they are ordered as the slopes in Section 5 imply: the more concentrated the stake, the larger the correction, with wealth, the most concentrated, moving the household sector furthest. The wealth estimate is the least precise because the wealth-weighted mean is effectively determined by a few hundred households (a Kish effective sample of about 4,600 after winsorizing, against 26,000 under equal weighting), but the wedge remains significant.

Table 3: The structural aggregators are significantly below the equal-weighted average

Aggregator	Channel	Sample	Wedge (pp)
Consumption	Aggregate demand	Michigan 1978–2025	-0.32 ($t = -31$)
Labor income	Wage Phillips curve	Michigan 1978–2025	-0.87 ($t = -15$)
Consumption	Aggregate demand	SCE 2013–2019	-0.24 ($t = -15$)
Labor income	Wage Phillips curve	SCE 2013–2019	-0.30 ($t = -7$)
Wealth	Bond pricing	SCE 2013–2019	-0.51 ($t = -12$)

Notes. Wedge is the structural aggregate minus the equal-weighted average. Standard errors from a bootstrap resampling survey months (Michigan) or households (SCE), 300–400 replications. Consumption is the Consumer Expenditure Survey expenditure schedule evaluated at the household's income rank; wealth is savings and investments, winsorized at the 99th percentile.

6.3. *How much of the gap to professional forecasters closes*

The correction closes a substantial and significant fraction of the gap that the literature reads as household pessimism. Over the months for which a professional forecast is available (1981–2025), the equal-weighted household average exceeds the Survey of Professional Forecasters median by 1.09 percentage points. Consumption weighting cuts this to 0.75 points and labor-income weighting to 0.62, so the consumption aggregator eliminates 31 percent of the household–professional gap and the labor-income aggregator 43 percent, purely by changing weights. This reduction is the same wedge that Table 3 shows is significant at $t = -31$, so the closing of the gap is real, not sampling noise.

Closing the gap to the professionals is not the same as being less accurate, and the post-2020 episode separates the two. In 2020–2024 the professional median averaged about 2.4 percent while realized inflation averaged about 4.5; the consumption-weighted household aggregate averaged close to realized inflation (Panel A of Figure 2). In the most consequential inflation episode in four decades the correctly weighted household sector forecast inflation more accurately than the professionals, and the aggregation correction moved the household aggregate toward, not away from, the realized outcome. This sharpens the policy reading of the level result: because the equal-weighted average sits above the consumption-weighted aggregate that governs demand, a central bank that tracks the average overstates how high the demand-relevant expectation is, may misread the gap as expectations de-anchoring, and risks tightening more aggressively than the correctly aggregated belief warrants.

6.4. Testing rationality

The construction in Section 2 makes no claim that the aggregate expectation is accurate; this subsection asks whether the correctly aggregated household belief is rational. The benchmark is full-information rational expectations, under which a forecast must be both *unbiased*, correct on average, and *efficient*, with errors uncorrelated with anything the forecaster knew when the forecast was made [Coibion & Gorodnichenko, 2015]. The household survey is usually judged to fail on both counts, read as too high and too slow to update, and these apparent failures motivate the structural belief distortions the literature builds into models [Carroll, 2003, Coibion & Gorodnichenko, 2015, Bhandari, Borovička & Ho, 2024]. Because the equal-weighted average is the object those verdicts were reached on, it is worth asking how much of the failure survives correct aggregation. I therefore subject each constructed aggregate to three standard tests, each isolating a different rationality property, and compare it to the professional benchmark. Throughout I treat the monthly aggregate as a one-year-ahead forecast of realized Consumer Price Index inflation and report Newey–West standard errors with eighteen lags, since the overlapping twelve-month horizon makes successive forecast errors serially correlated.

Unbiasedness. Let F_t be the aggregate forecast made in month t, let π^r_{t+1} be the realized twelve-month inflation it targets, and define the forecast error $e_t \equiv F_t - \pi^r_{t+1}$. The first test asks whether the error is zero on average:

$$H_0: \ \mathbb{E}[e_t] = 0, \qquad e_t \equiv F_t - \pi^r_{t+1}, \tag{35}$$

estimated as the mean error. This is exactly the property the conventional reading disputes: the claim that households are "too pessimistic" is the claim that $\mathbb{E}[e_t] > 0$. It is also the weakest of the three, because a forecast can be unbiased on average yet wrong in systematic, predictable ways.

Mincer–Zarnowitz optimality. The second test regresses realized inflation on the forecast [Mincer & Zarnowitz, 1969]:

$$\pi^r_{t+1} = a + bF_t + \varepsilon_t, \qquad H_0: \ a = 0 \text{ and } b = 1. \tag{36}$$

An optimal forecast satisfies the joint null: no constant bias ($a = 0$) and a one-for-one relationship with what it predicts ($b = 1$), so the forecast neither damps nor amplifies the variation in realized inflation. Mincer–Zarnowitz is therefore not a third, separate property but a joint test that bundles the other two: the intercept condition $a = 0$ is a level, or bias, requirement, and the slope condition $b = 1$ is a scaling, or efficiency, requirement. It is accordingly stricter than the mean-bias test, since a forecast can have a zero mean error yet systematically over- or under-scale movements in inflation, which the slope b detects; I report the joint F-statistic for (36).

Efficiency. The third test asks whether the error could have been predicted from information already in the forecaster's hands, the defining requirement of rational expectations. I regress the error on the forecast itself, the simplest piece of that information:

$$e_t = c + d\,F_t + u_t, \qquad H_0:\ d = 0. \tag{37}$$

A nonzero d means the error is predictable from the forecast, so a forecaster who shaded toward ($d > 0$) or away from ($d < 0$) her own forecast would have done better; a significantly positive d is the signature of under-reaction, sluggish updating, and a negative d of over-reaction. This is the weak-form counterpart of the information-rigidity test of Coibion & Gorodnichenko [2015], who regress the consensus error on the consensus forecast *revision* to gauge how sluggishly forecasters absorb news; using the forecast level F_t rather than the revision asks the same question, whether errors are orthogonal to time-t information, with the information the rolling survey makes readily available. A full revision-based test, which needs fixed-target forecast revisions the rolling survey does not cleanly provide, is a natural complement. This slope condition is mechanically the Mincer–Zarnowitz slope condition: rearranging (36) gives $e_t = -a + (1-b)F_t$, so $d = 1-b$ and the efficiency null $d = 0$ coincides with $b = 1$, while Mincer–Zarnowitz differs only by also imposing the level condition $a = 0$. This is why, in Table 4, the professionals fail Mincer–Zarnowitz through the slope (they are inefficient, $b \neq 1$) whereas the household aggregates fail it through the intercept (they are biased, $a \neq 0$, but efficient).

Table 4 reports the three tests for the Michigan sample. Correct aggregation removes a substantial part of the apparent irrationality without eliminating it. On the first two tests the aggregates improve but do not pass: the mean bias falls from 0.94 percentage points for the average to 0.63 for the consumption aggregate and 0.51 for the labor-income aggregate, and the Mincer–Zarnowitz statistic falls from 13.6 to 7.8 and 5.9, yet all continue to reject, so the aggregates are less biased than the average but not unbiased. The professional benchmark fails the *third* test instead: the Survey of Professional Forecasters is essentially unbiased, with a mean error of 0.10 percentage points, but its errors are strongly predictable from its own forecast (efficiency slope 0.54, $t = 3.7$), the under-reaction, or information rigidity, that Coibion & Gorodnichenko [2015] document for professional forecasters. The household aggregates pass this efficiency test: their errors are not predictable from the forecast. Households and professionals are thus irrational in different ways, households biased but efficient and professionals unbiased but inefficient, and correct aggregation narrows the household bias toward the professional one while preserving the household efficiency advantage.

What do the three tests together say about full-information rational expectations? It helps to split FIRE into its two ingredients, as Coibion & Gorodnichenko [2015] do, because the two forecasters fail on different ones. The professional forecast departs from *full information*: its errors are predictably slug-

Table 4: Rationality tests, Michigan one-year inflation expectations, 1978–2025

	Bias (pp) [mean error]	Mincer–Zarnowitz F: a=0, b=1	Efficiency slope [error on forecast]
Average (equal-weighted)	+0.94	13.6	−0.14
	(3.7)	[0.00]	(−0.7)
Aggregate (consumption)	+0.63	7.8	−0.17
	(2.6)	[0.00]	(−0.9)
Aggregate (labor income)	+0.51	5.9	−0.16
	(2.2)	[0.00]	(−1.0)
Survey of Professional Forecasters	+0.10	9.2	+0.54
	(0.4)	[0.00]	(3.7)

Notes. t-statistics in parentheses, p-values in brackets, Newey–West standard errors with eighteen lags. Bias is the mean of (forecast − realized). The Mincer–Zarnowitz column reports the F-statistic for the joint hypothesis $a = 0, b = 1$ in $\text{realized}_t = a + b\,\text{forecast}_t$. The efficiency column reports the slope of (forecast − realized) on the forecast.

gish, the under-reaction that sticky- or noisy-information models generate, yet it is unbiased, so rational expectations itself is hard to reject for the professionals, which is exactly CG's reading of the consensus. The correctly aggregated household belief departs on the other margin. Its errors are not predictable from the forecast, so it shows none of the professionals' information rigidity on this test; what it retains is a level bias of about half a percentage point, which the unbiasedness and Mincer–Zarnowitz tests still reject. Correct aggregation roughly halves that bias relative to the equal-weighted average, stripping out the part that was an artifact of weighting, but does not erase it. The verdict is therefore not that the aggregate is fully rational, but something more precise: its residual departure from FIRE is a modest level bias rather than information rigidity, it is far smaller than the equal-weighted average implies, and on the efficiency dimension the household aggregate is at least as well behaved as the professional benchmark. Whether the remaining bias reflects genuine over-prediction, asymmetric loss, or differences in the inflation households still face is something these tests cannot separate.

This reading bears directly on the use of survey expectations as structural inputs. Bhandari, Borovička & Ho [2024] discipline subjective beliefs in a business-cycle model with exactly this kind of household-survey evidence, both the time-series and the cross-sectional pessimism, and find the resulting belief distortions quantitatively large: pessimism raises inflation and unemployment forecasts and, working through the aggregate-demand channel, lowers economic activity. My results speak to the size of that distortion on the inflation dimension. The cross-sectional pessimism they feed in, that low-income, low-spending households expect higher inflation, is precisely the dispersion that the aggregation correction nets out for the demand-relevant object: by Proposition 2 the belief that enters the aggregate Euler equation is the consumption-weighted aggregate, which is materially less pessimistic than the equal-weighted average. The very channel through which their pessimism operates, aggregate demand, is the one where reweighting most reduces measured pessimism, so a meaningful share of what an equal-weighted reading attributes to a structural, demand-suppressing belief distortion is an artifact of aggregation rather than a distortion of the belief that governs demand. This does not overturn their mechanism, which also runs through unemployment and income beliefs I do not measure, but it implies that the inflation pessimism relevant for aggregate demand is smaller than equal-weighted survey inputs suggest, and that the same reweighting should be applied before cross-sectional survey dispersion is read as a structural distortion.

7. Conclusion

The empirical literature reads the pessimism of equal-weighted household inflation surveys as evidence of irrationality. This paper offers a different reading: much of that pessimism is an artifact of the aggregator. The equal-weighted average is the expected inflation of a price index in which every household basket counts the same, an index no one publishes; the structurally consistent object weights each household by its expenditure share, exactly as the Bureau of Labor Statistics builds the Consumer Price Index. The two differ by the cross-sectional covariance between expenditure and expected inflation, which is large and negative, so correct aggregation makes the household sector look markedly less pessimistic and less irrational.

The central result is that this weight is not merely an accounting convention. In an otherwise standard heterogeneous-agent New Keynesian model perturbed only by heterogeneous inflation beliefs, the consumption share is exactly the weight on household beliefs in the aggregate Euler equation, so the measure that reconciles the survey with the published index is the measure that governs aggregate demand. Two implications follow. When central banks cite household expectations in deliberations on anchoring, the consumption-share aggregate, not the equal-weighted average, is the object to compare with realized inflation or the target; reading the average instead overstates the demand-relevant belief and risks tightening more than it warrants. And Phillips-curve regressions that feed in the equal-weighted average use a measurement that matches no structural object: the appropriate aggregator is object-dependent, consumption-share weighting for aggregate demand, labor-income-share weighting for the wage Phillips curve, and wealth-share weighting for bond pricing, and the equal-weighted average is none of them.

A. Proof of Proposition 1

Let $\bar{s} \equiv \int_0^1 s_i\, di$ and $\bar{F} \equiv \int_0^1 F_i\, di = \bar{\pi}_t^{\text{avg}}$. The normalization $s_i = E_i/\bar{E}$ gives $\bar{s} = 1$, so $s_i - 1 = s_i - \bar{s}$. Taking the difference of the two definitions and combining the integrals,

$$\bar{\pi}_t^{\text{agg}} - \bar{\pi}_t^{\text{avg}} = \int_0^1 s_i F_i\, di - \int_0^1 F_i\, di = \int_0^1 (s_i - \bar{s})\, F_i\, di.$$

Because $\int_0^1 (s_i - \bar{s})\, di = 0$ and $\bar{F}$ is constant, the term $\int_0^1 (s_i - \bar{s})\, \bar{F}\, di$ is zero; subtracting it leaves the value unchanged and completes the centered product,

$$\int_0^1 (s_i - \bar{s})\, F_i\, di = \int_0^1 (s_i - \bar{s})(F_i - \bar{F})\, di = \text{Cov}_i(s_i, F_i).$$

Hence $\bar{\pi}_t^{\text{agg}} - \bar{\pi}_t^{\text{avg}} = \text{Cov}_i(s_i, F_i)$, and multiplying by -1 yields (6). $\square$

B. Relaxing the own-basket forecasting assumption

Proposition 1 is stated under Assumption 1, which equates each household's reported forecast F_i with its own-basket forecast $\tilde{F}_i \equiv E_i^t[\pi_{i,t+1}]$. This appendix dispenses with that assumption, allowing a household's stated forecast of inflation in general to differ from the forecast implied by its own consumption basket. Throughout, the conventional measure remains the equal-weighted average of the reported forecasts, $\bar{\pi}_t^{\text{avg}} \equiv \int_0^1 F_i\,di$, and the structurally consistent object is the consumption-share-weighted aggregate of the own-basket forecasts that the identity (3) requires, $\bar{\pi}_t^{\text{agg}} \equiv \int_0^1 s_i \tilde{F}_i\,di$, with $s_i \equiv E_i/\bar{E}$ and $\int_0^1 s_i\,di = 1$. The following proposition generalizes Proposition 1.

Proposition 4 (Average versus aggregate, general case)**.** *Without Assumption 1, the average and aggregate expectations differ by*

$$\bar{\pi}_t^{avg} - \bar{\pi}_t^{agg} = \underbrace{-\,\text{Cov}_i(s_i, F_i)}_{weighting} + \underbrace{\int_0^1 s_i\,(F_i - \tilde{F}_i)\,di}_{within\text{-}respondent\ gap}. \tag{38}$$

The first term is the consumption-share–belief covariance of Proposition 1; the second is the consumption-weighted average, within respondents, of the gap between the reported and the own-basket forecast. Proposition 1 is the special case $F_i = \tilde{F}_i$ for all i, in which the second term vanishes.

Proof. Add and subtract $\int_0^1 s_i F_i\,di$ and use $\int_0^1 s_i\,di = 1$:

$$\begin{aligned}
\bar{\pi}_t^{\text{avg}} - \bar{\pi}_t^{\text{agg}} &= \int_0^1 F_i\,di - \int_0^1 s_i \tilde{F}_i\,di \\
&= \underbrace{\left(\int_0^1 F_i\,di - \int_0^1 s_i F_i\,di\right)}_{=-\int_0^1 (s_i-1)F_i\,di} + \left(\int_0^1 s_i F_i\,di - \int_0^1 s_i \tilde{F}_i\,di\right) \\
&= -\,\text{Cov}_i(s_i, F_i) + \int_0^1 s_i\,(F_i - \tilde{F}_i)\,di,
\end{aligned}$$

where the final line uses $\int_0^1 (s_i - 1)F_i\,di = \int_0^1 (s_i - \bar{s})(F_i - \bar{F})\,di = \text{Cov}_i(s_i, F_i)$ with $\bar{s} = 1$, exactly as in the proof of Proposition 1. $\square$

Corollary 1 (Lower bound)**.** *If the reported forecast weakly exceeds the own-basket forecast for almost every household, $F_i \geq \tilde{F}_i$, then*

$$\bar{\pi}_t^{avg} - \bar{\pi}_t^{agg} \;\geq\; -\,\text{Cov}_i(s_i, F_i), \tag{39}$$

with equality under Assumption 1. The covariance of Proposition 1 is then a lower bound on the full gap between the conventional average and the structurally consistent aggregate, whatever the degree of own-basket loading.

Proof. Because $s_i \geq 0$, the hypothesis $F_i \geq \tilde{F}_i$ implies $\int_0^1 s_i (F_i - \tilde{F}_i)\, di \geq 0$, so the within-respondent term in (38) is non-negative; equality holds when $F_i = \tilde{F}_i$ almost everywhere, which is Assumption 1. □

The hypothesis of Corollary 1 has direct empirical support: Dietrich et al. [2023] document that a respondent's stated expectation of overall inflation exceeds the basket-weighted average of her own category-level expectations, that is $F_i \geq \tilde{F}_i$, so the lower bound applies in the data.

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